A Non-Uniform Crossnorm Partial Answer to an Unconditional-Basis Tensor Product Problem

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Source and Status

This packet concerns the tensor-norm question in:

> Rafik Karkri and Samir Kabbaj, *A New Crossnorm That Preserves Unconditional Bases in Banach Spaces*, arXiv:2506.22522.

The source paper frames the original problem as the existence of a tensor norm, that is, a uniform reasonable crossnorm, preserving unconditional Schauder bases for all Banach-space tensor products. Page 1 also states that this uniform existence question remains open, while the paper constructs a generally non-uniform crossnorm for each fixed pair.

Abstract

Let α be a tensor norm (i.e., a uniform reasonable crossnorm) on the class of all algebraic tensor products of Banach spaces $E \otimes F$. We say that α preserves unconditionality if, for every pair of Banach spaces E and F with unconditional Schauder bases (USBs), the completion $E \otimes_\alpha F$ also admits a USB.

It is well known that none of Grothendieck's fourteen natural tensor norms satisfy this unconditionality-preserving condition. Moreover, the existence of a tensor norm α with this property remains an open question.

In this paper, we construct for every such pair (E, F) a new reasonable crossnorm α. This norm has the surprising property that—despite being generally non-uniform—the space $E \otimes_\alpha F$ nevertheless admits a USB.

Keywords: Frame, Basis, Besselian, Unconditional, Tensor protuct.
2008 MSC: 46B04, 46B15, 46B25, 46B45, 46A32

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1 Introduction

Literature-Implied Partial Result

Theorem 1. *Let E and F be Banach spaces equipped with unconditional Schauder bases*

$$\mathcal{F} = ((a_m, b_m^*))_{m\geq 1}, \qquad \mathcal{G} = ((x_n, y_n^*))_{n\geq 1}.$$

For this fixed pair of spaces and chosen bases, the source paper constructs equivalent Besselian norms on E and F and a pair-and-basis-dependent crossnorm

$$\alpha_{\mathcal{F},\mathcal{G}}^{\mathrm{Bess}}$$

on $E \otimes F$ such that:

1. *$\alpha_{\mathcal{F},\mathcal{G}}^{\mathrm{Bess}}$ is a reasonable crossnorm on the tensor product of the corresponding Besselian renormings, and*
$$\varepsilon_{\mathcal{F},\mathcal{G}}(u) \leq \alpha_{\mathcal{F},\mathcal{G}}^{\mathrm{Bess}}(u) \leq \pi_{\mathcal{F},\mathcal{G}}(u) \qquad (u \in E \otimes F).$$

2. *The square-ordered tensor product sequence*
$$\mathcal{F} \otimes \mathcal{G} = ((a_m \otimes x_n, b_m^* \otimes y_n^*))_{m,n\geq 1}$$
forms an unconditional Schauder basis of
$$E \otimes_{\alpha_{\mathcal{F},\mathcal{G}}^{\mathrm{Bess}}} F.$$

Proof. This is a source-backed literature-status statement rather than an independent new proof.

The construction of $\alpha_{\mathcal{F},\mathcal{G}}^{\mathrm{Bess}}$ is given in the source paper by the definition preceding Proposition `New-crossnorm`. That proposition proves that, after passing to the equivalent Besselian norms on E and F, the constructed functional is a reasonable crossnorm and satisfies the injective/projective bounds displayed above.

When $\mathcal{F}$ and $\mathcal{G}$ are unconditional Schauder bases, the corresponding source theorem applies directly; in the source file it is labelled `tensor.bess.is.bess`. It states that the tensor product sequence, ordered by square ordering, forms an unconditional Schauder basis in the completion for $\alpha_{\mathcal{F},\mathcal{G}}^{\mathrm{Bess}}$. This is exactly the asserted non-uniform preservation statement. □

Why This Is Only Partial

The original problem asks for a tensor norm in the usual sense: a uniform reasonable crossnorm defined functorially across Banach-space tensor products. The Besselian crossnorm above depends on the particular pair (E, F) and on the chosen bases $(\mathcal{F}, \mathcal{G})$. The source paper also states and demonstrates that this crossnorm is generally not uniform.

Therefore the packet records a strong partial answer:

pair/basis-dependent reasonable crossnorm preserving USBs,

not a full solution of the uniform tensor-norm existence problem.

Review Checklist

- Check that the downstream problem accepts equivalent Besselian renormings; otherwise the result should be cited with this renorming qualification.
- Check that the term "tensor norm" is not used for this construction without the non-uniform caveat.
- Treat this packet as a literature-status partial result unless a separate argument upgrades the construction to a uniform tensor norm.